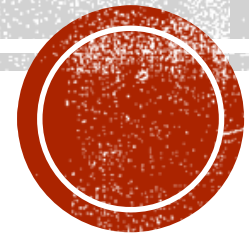


EEE141L: ELECTRICAL CIRCUITS LAB

LAB: 08

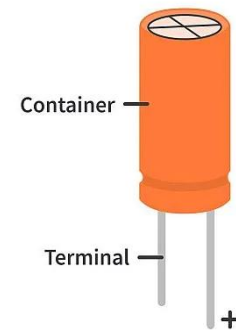
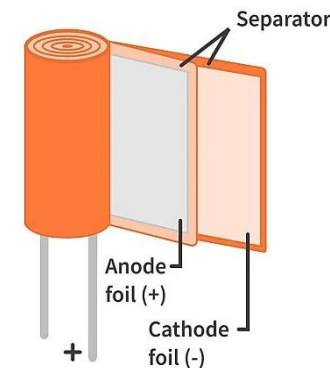
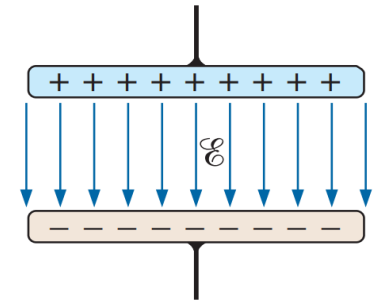
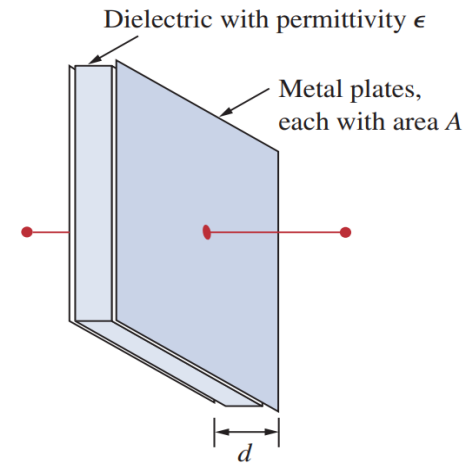
EXPERIMENT NAME: CAPACITOR CHARGING AND DISCHARGING IN RC
CIRCUITS

LAB INSTRUCTOR: SUDIPTO SARKER SUPTO



CAPACITORS

- A capacitor is a passive element designed to store energy in its electric field, commonly for a backup release of energy if the power fails.
- Capacitors do this by storing electric charge, which creates a build up of electric potential energy.
- They are made in the form of two conductive metal plates connected to a voltage supply (parallel plate capacitor).
- There is commonly a dielectric in between the plates, to ensure charge does not flow across them.



CAPACITANCE

- Capacitors are marked with a value of their **capacitance**.
- Capacitance is a measure of a capacitor's ability to store charge on its plates—in other words, its storage capacity.
- In other words, capacitance is defined as the charge stored per unit potential difference (between the plates).

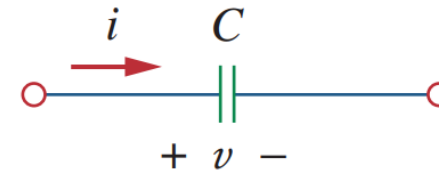
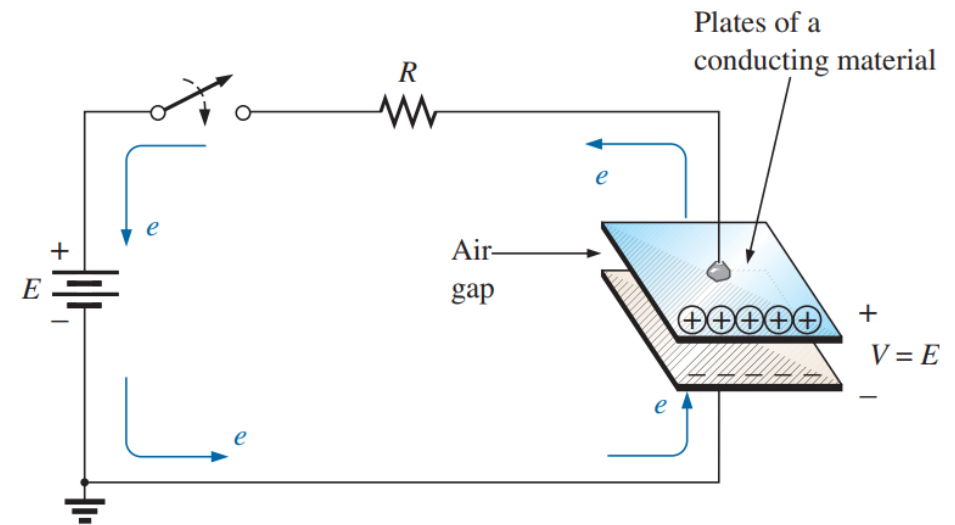
$$C = \frac{Q}{V}$$

C = farads (F)

Q = coulombs (C)

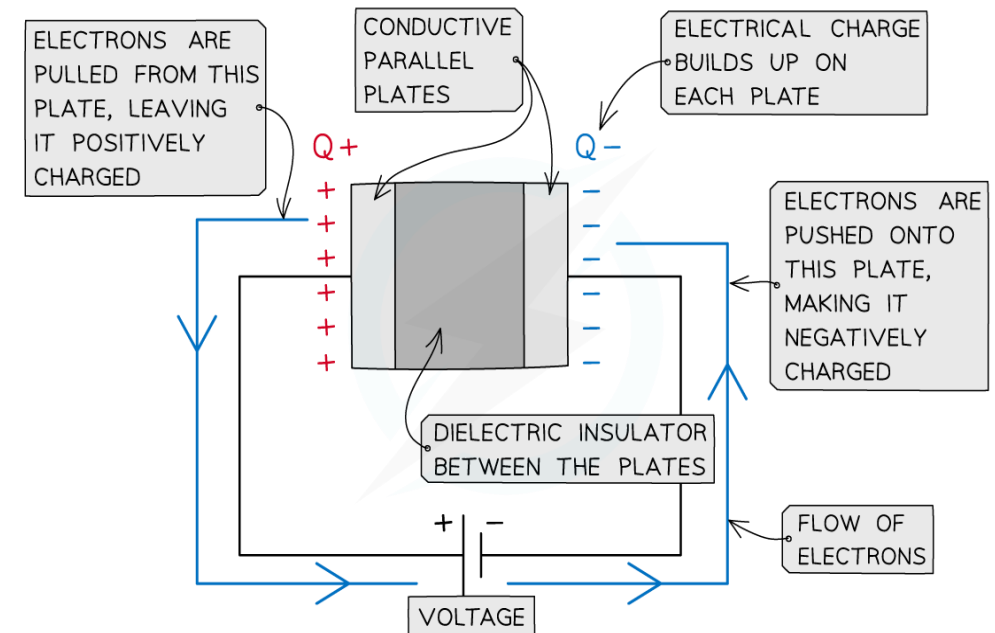
V = volts (V)

- The higher the capacitance of a capacitor, the greater is the amount of charge stored on the plates for the same applied voltage.



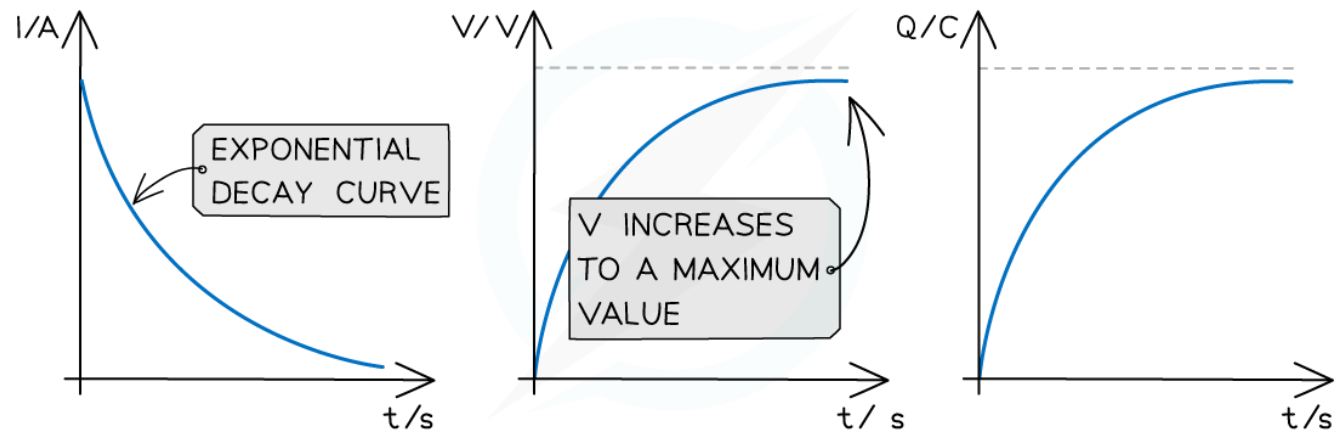
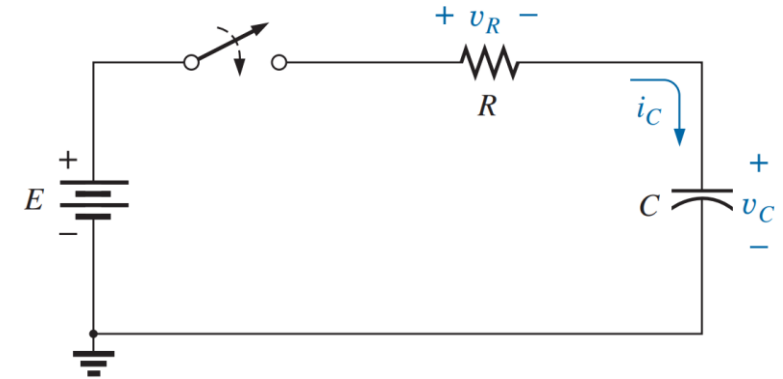
CHARGING OF A CAPACITOR

- Capacitors are charged by a power supply (e.g. a battery).
- When charging, the electrons are pulled from the plate connected to the positive terminal of the power supply. Hence the plate nearest the positive terminal is **positively charged**.
- They travel around the circuit and are pushed onto the plate connected to the negative terminal. Hence the plate nearest the negative terminal is **negatively charged**.
- As the negative charge builds up, fewer electrons are pushed onto the plate due to electrostatic repulsion from the electrons already on the plate. When no more electrons can be pushed onto the negative plate, the charging stops.



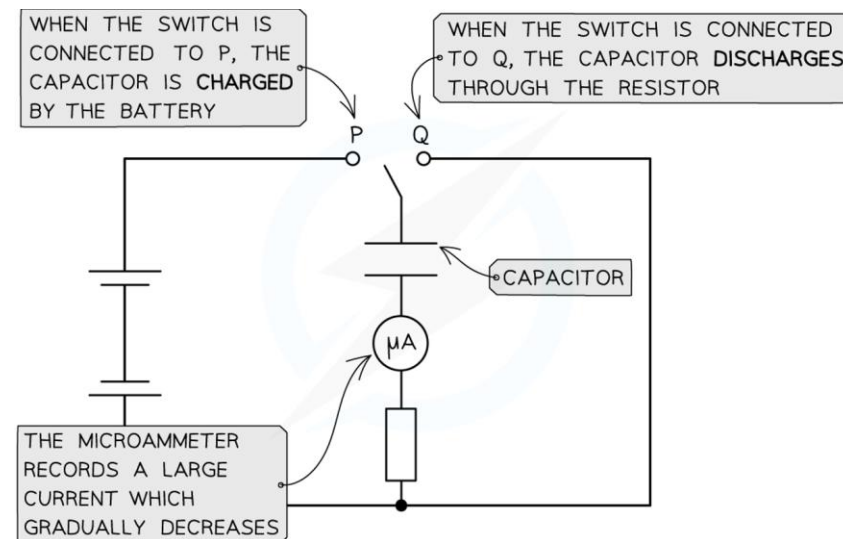
BASIC RC CHARGING CIRCUIT

- At the start of charging, the current is large and gradually falls to zero as the electrons stop flowing through the circuit. The current decreases exponentially.
- Since an equal but opposite charge builds up on each plate, the potential difference between the plates slowly increases until it is the same as that of the power supply.



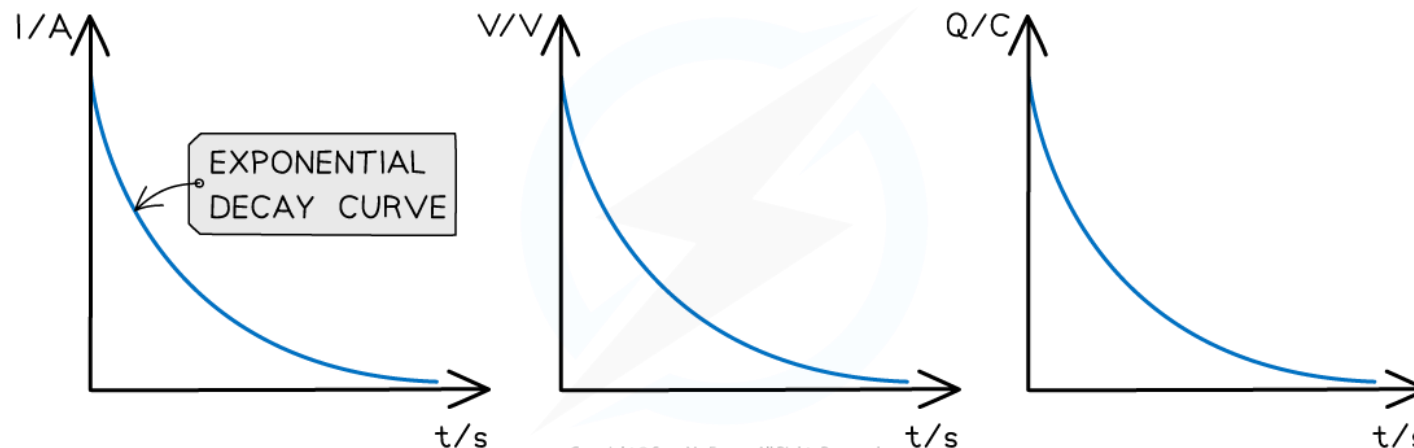
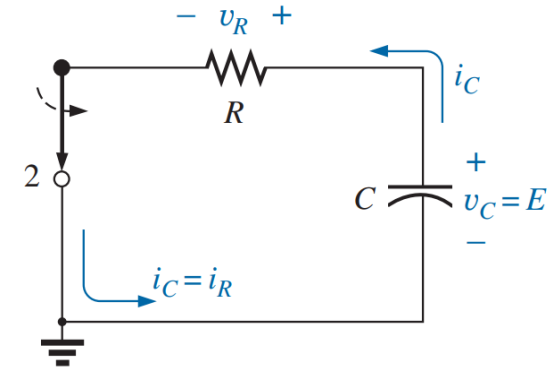
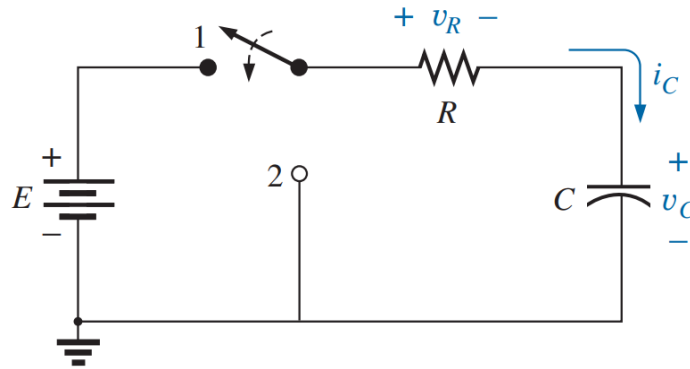
DISCHARGING OF A CAPACITOR

- Capacitors are discharged through a resistor with no power supply present.
- The electrons now flow back from the negative plate to the positive plate until there are equal numbers on each plate and no potential difference between them.
- Charging and discharging is commonly achieved by moving a switch that connects the capacitor between a power supply and a resistor.



BASIC RC DISCHARGING CIRCUIT

- At the start of discharge, the current is large (but in the opposite direction to when it was charging) and gradually falls to zero.
- As a capacitor discharges, the current, p.d. and charge all decrease exponentially.



TIME CONSTANT (τ)

- The time constant (τ) of a capacitor discharging through a resistor is a measure of how long it takes for the capacitor to discharge.
- The definition of the time constant is: **The time taken for the charge, current or voltage of a discharging capacitor to decrease to 37% of its original value.**
- Alternatively, for a **charging** capacitor: **The time taken for the charge or voltage of a charging capacitor to rise to 63% of its maximum value.**
- The quantity τ is defined by:

$$\tau = RC \quad (\text{time, s})$$

- The time to half, $t_{1/2}$ (half-life) for a discharging capacitor is: **The time taken for the charge, current or voltage of a discharging capacitor to reach half of its initial value.** [$t_{1/2} = \ln(2)\tau \approx 0.69\tau = 0.69RC$]

RC CHARGING CIRCUIT CURVE

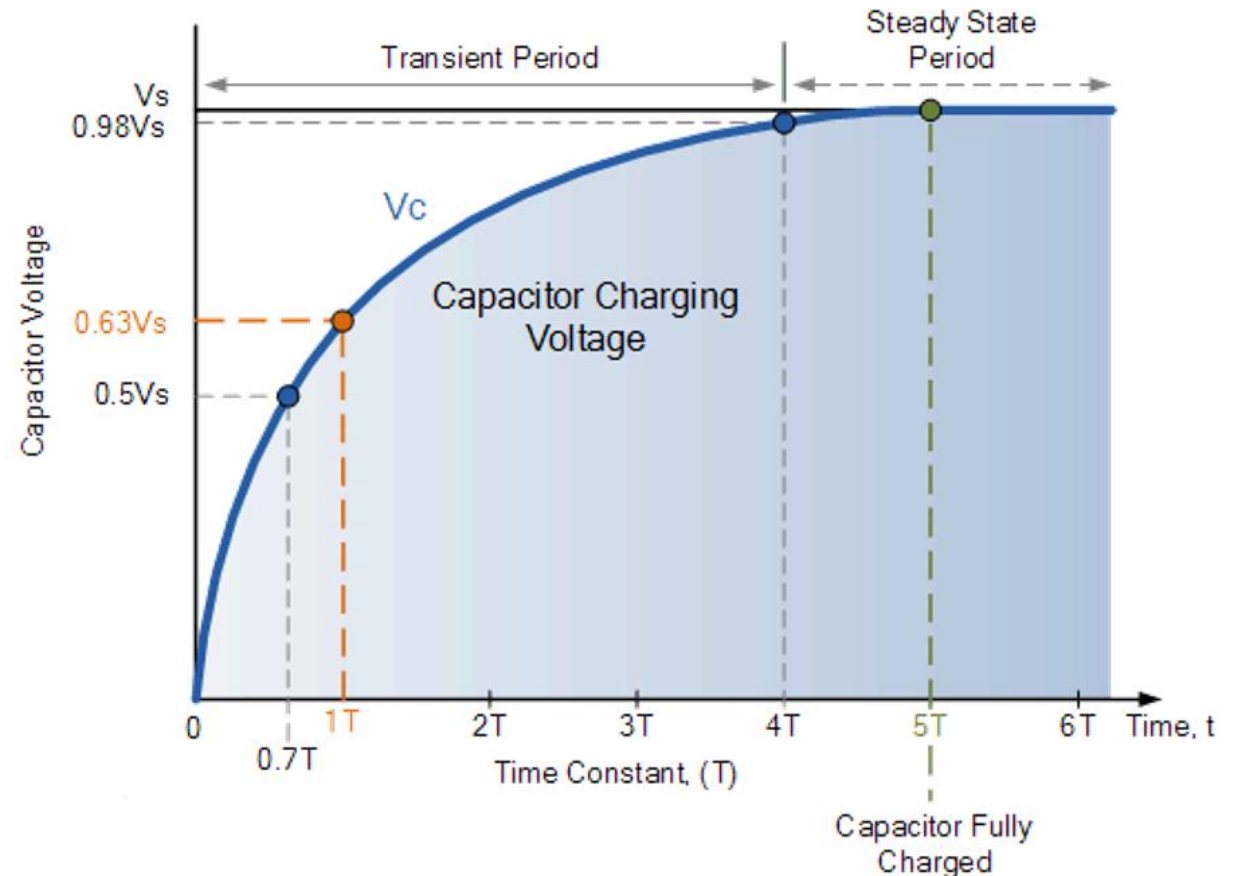
- The voltage across a capacitor in a dc network is essentially equal to the applied voltage after five time constants of the charging phase have passed.

$$v_C = E(1 - e^{-t/\tau}) \quad \text{(volts, V)} \quad \text{charging}$$

- The current of a capacitive dc network is essentially zero amperes after five time constants of the charging phase have passed.

$$i_C = \frac{E}{R} e^{-t/\tau} \quad \text{(amperes, A)} \quad \text{charging}$$

- During the charging phase, the major change in voltage and current occurs during the first time constant.



RC DISCHARGING CIRCUIT CURVE

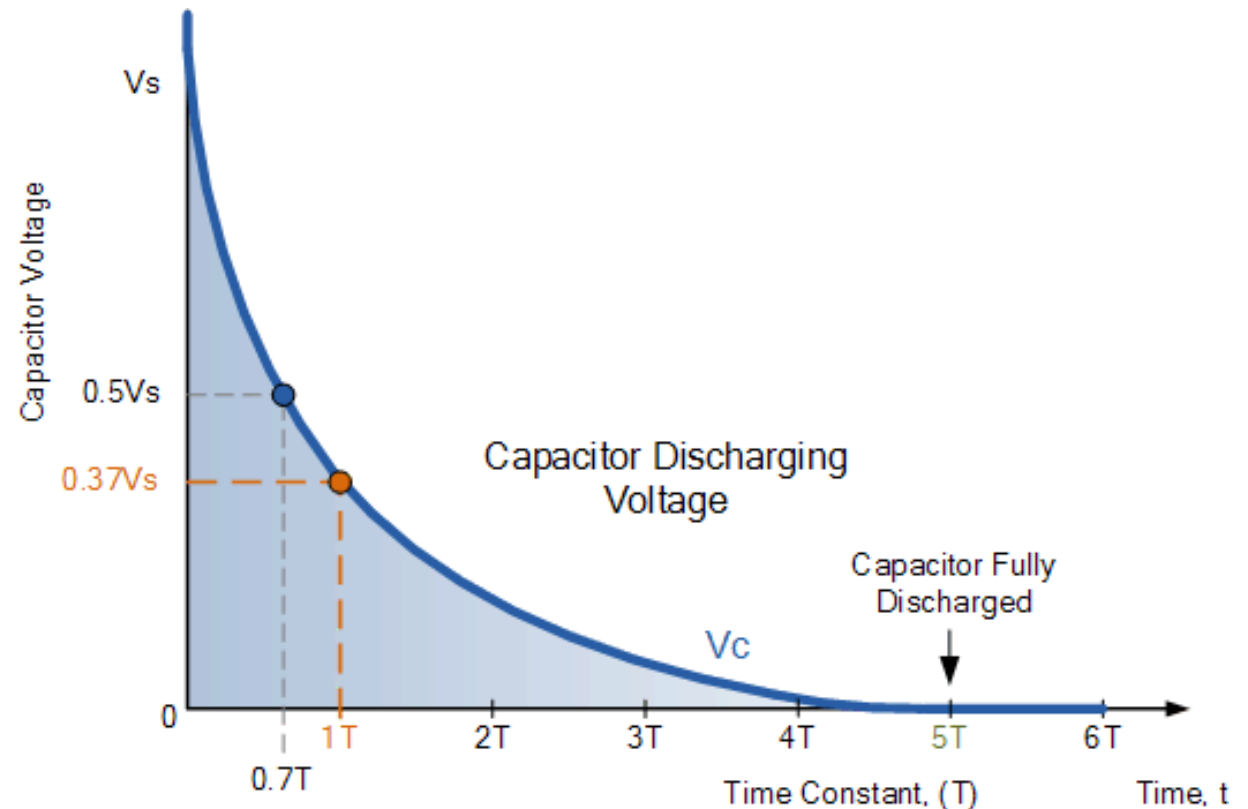
- The voltage across a capacitor in a dc network is essentially zero volts after five time constants of the discharging phase have passed. The expression is:

$$v_C = Ee^{-t/\tau} \quad \text{(volts, V)} \quad \text{discharging}$$

- The current of a capacitive dc network is essentially zero amperes after five time constants of the discharging phase have passed.

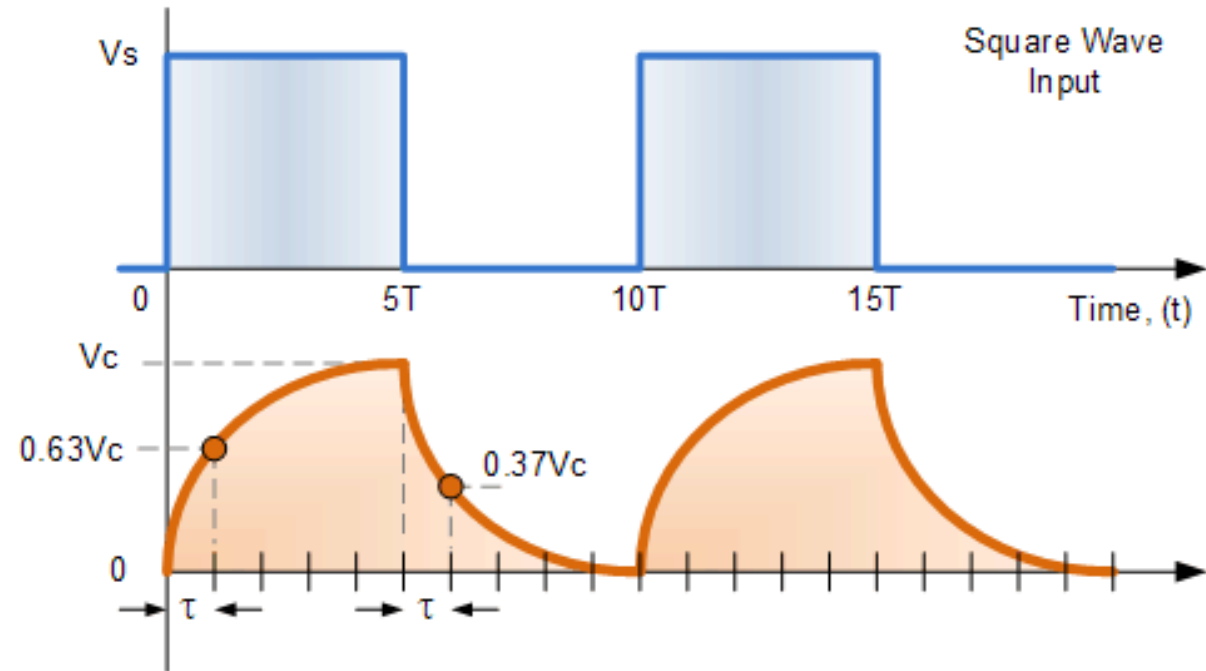
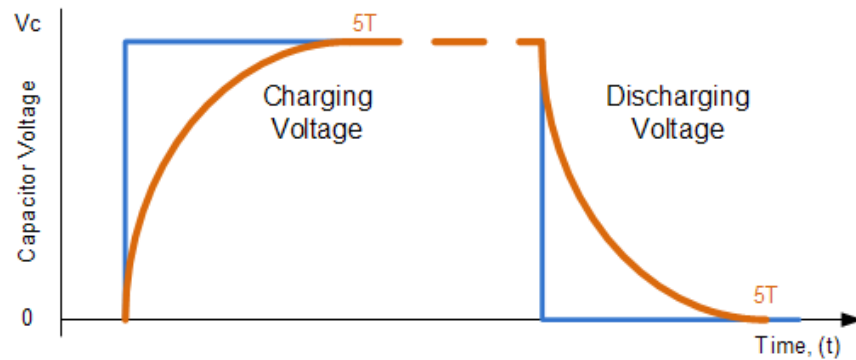
$$i_C = \frac{E}{R}e^{-t/\tau} \quad \text{(amperes, A)} \quad \text{discharging}$$

- During the discharging phase, the major change in voltage and current occurs during the first time constant.

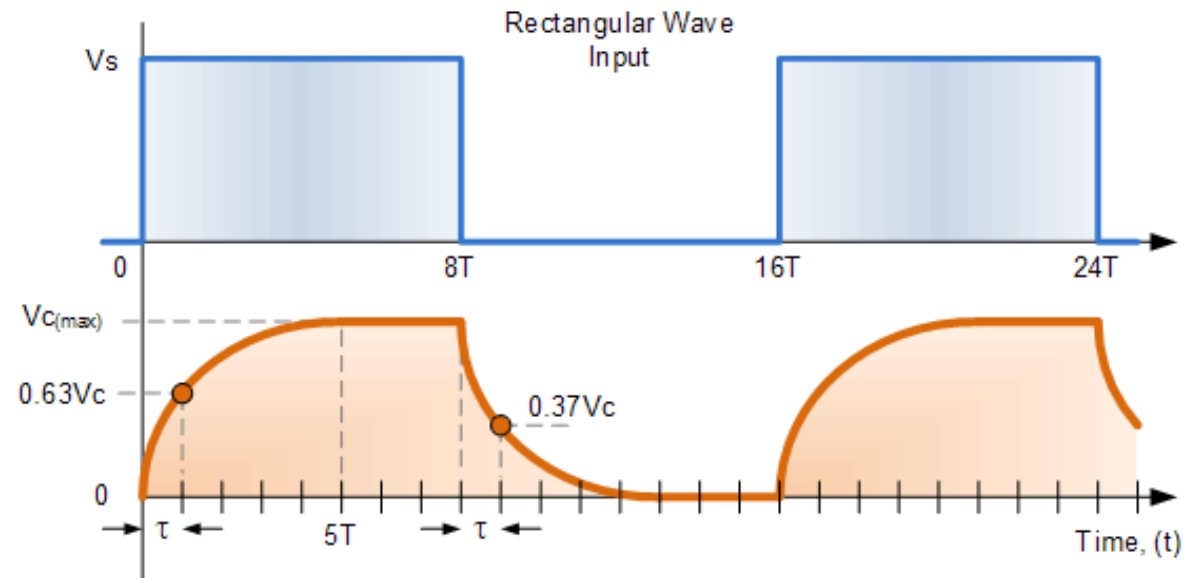
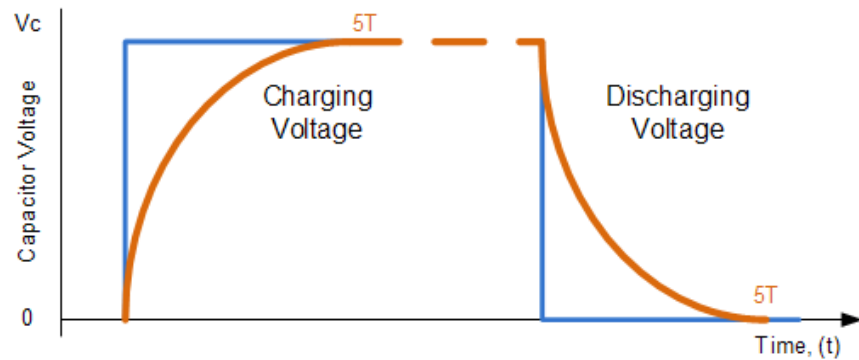


TYPICAL RC WAVEFORMS:

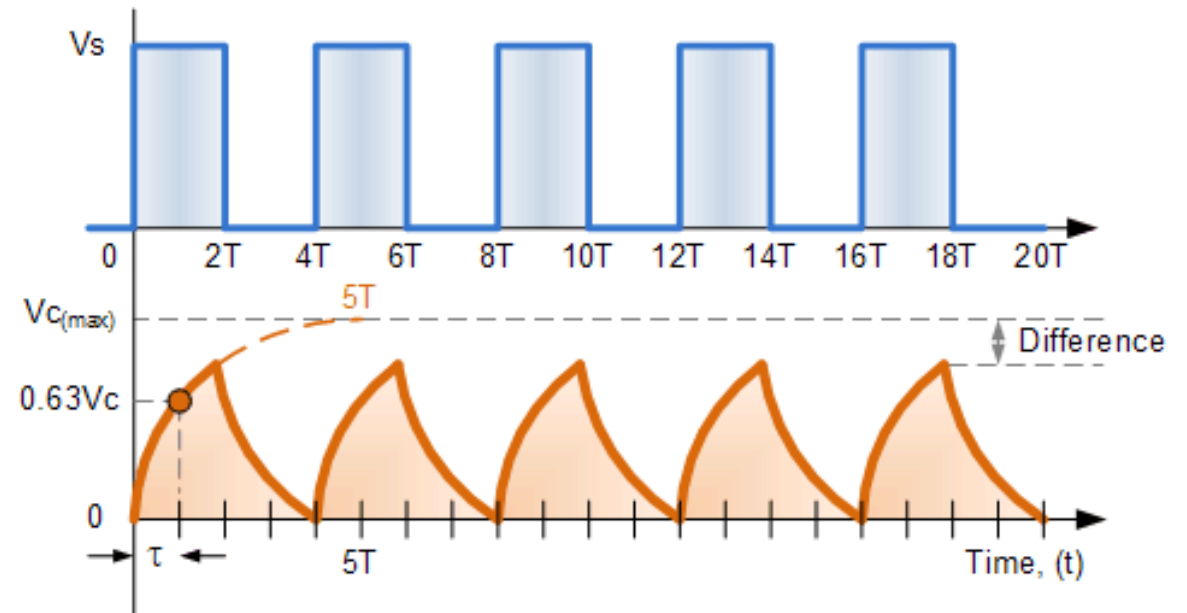
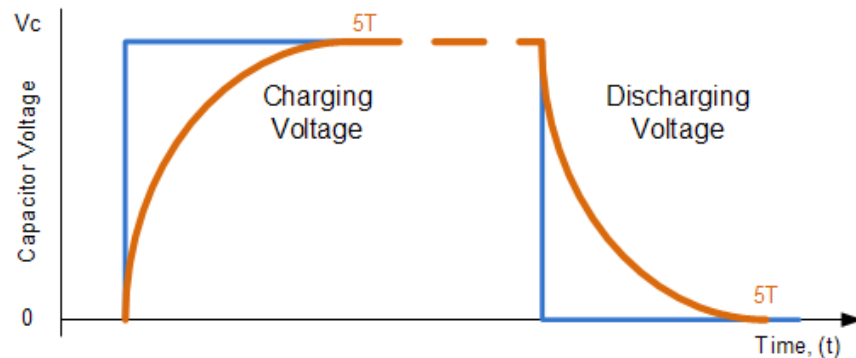
EXACT RC WAVEFORMS OF 5RC



TYPICAL RC WAVEFORMS: A LONGER 8RC INPUT WAVEFORM



TYPICAL RC WAVEFORMS: A SHORTER 4RC INPUT WAVEFORM



THE END...

Prepared by: Sudipto Sarker Supto

